Improved Comparative Phenotype Searches Between Human and Mouse at the HMDC

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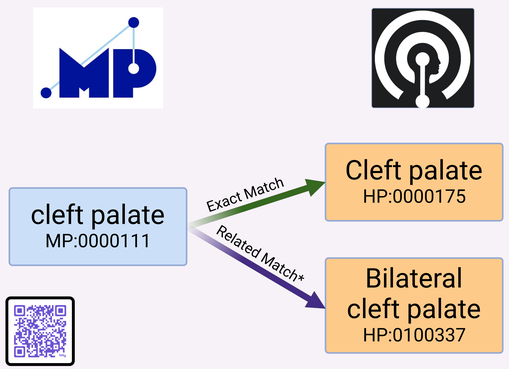
ABSTRACT

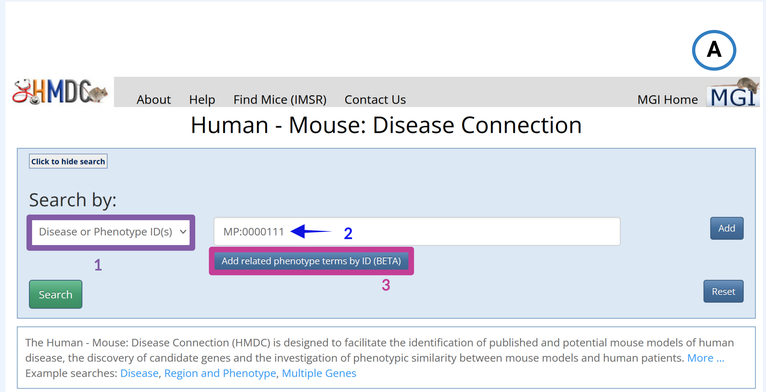
The Human - Mouse Disease Connection (HMDC; https://www.informatics.jax.org/humandisease.shtml) search tool at Mouse Genome Informatics (MGI) allows users to find and compare mutant phenotype and disease associations for human and mouse genes. Users may search by gene, phenotype or disease by name, synonym or ID and/or by genome region. Searching by ID allows for users to select a precise search term. Searching by a phenotype ID restricts results to a single species because human data are annotated using Human Phenotype (HP) ontology terms while mouse data are annotated using Mammalian Phenotype (MP) ontology terms. To help users find IDs for the corresponding terms in each ontology MGI has developed an MP-HP term match tool. This tool uses manually curated mappings between MP and HP terms, primarily created by MGI, publicly available in SSSOM formatted files. These mappings are supplemented with lexical mappings using term label and synonym matches. Only precise text matches are used for the lexical mappings. Users may search by one or more phenotype IDs and find potential counterparts in the other phenotype ontology. The results returned include the match method and type, matched term label and synonyms, and matched term definition. For example, a search for MP:0000111 (cleft palate) returns two HP terms; HP:0000175 (cleft palate, exact match) and HP:0100337 (Bilateral cleft palate, related match). Users may select one or more of the matched terms to add to the HMDC search. Searching the HMDC using these IDs limits the results to phenotypes. A similar search for 'cleft palate' using the phenotype or disease name search field also includes results for disease term names that include 'cleft palate'. Thus this new tool will improve the precision of searches in the HMDC by allowing users greater control using matched terms. Supported by NHGRI grant HG000330

SSSOM

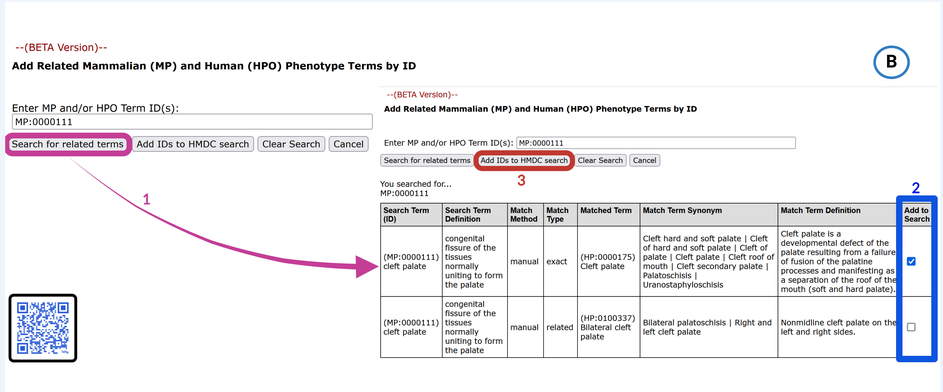


Mappings between MP and HPO terms are made using the Simple Standard for Sharing Ontology Mappings (SSSOM). Mappings are done manually by MP curators or algorithmically by IMPC or MGI. A single term in either ontology may have more than 1 mapping to terms in the other ontology. *The mapping between *cleft palate* and *Bilateral cleft palate* is based on an older mapping of *cleft palate* and the now obsolete term *Bilateral cleft lip and palate*. Because the new term has the ID for the obsolete term as an alternate ID the mapping has been transferred to the new term.

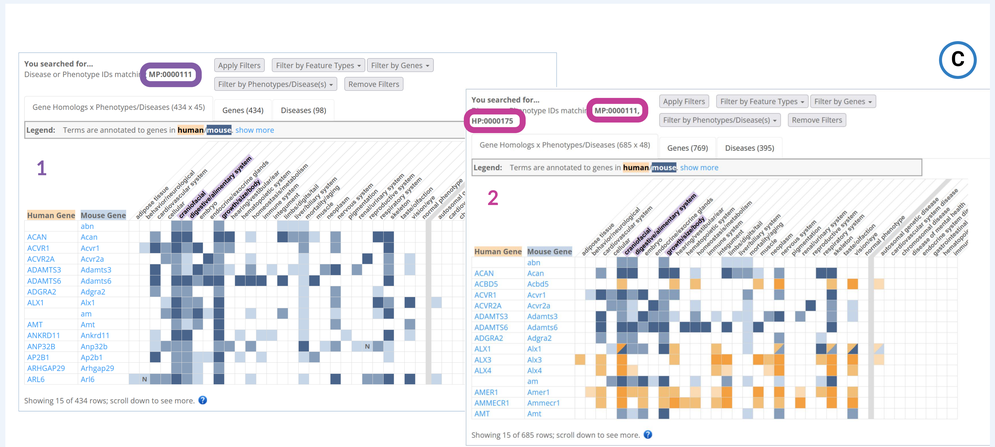
Using the MP-HP Match Tool



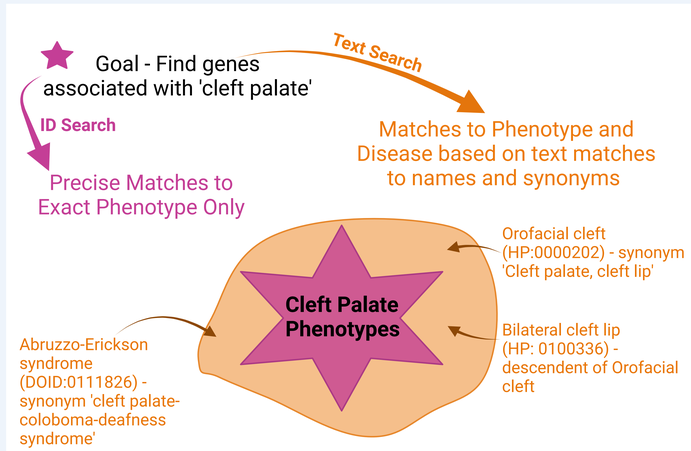
HMDC home page. To get to the MP-HP Match Tool (1) Select Disease or Phenotype ID(s) under Search by. Then (2) enter the ID(s) you want to search by in the box and (3) click Add related terms by ID (BETA) under the search box. This will launch the search tool with the ID(s) in the search box already filled in.



MP-HP Match Tool: In the tool any IDs entered in the HMDC search are displayed in the search box. You may also add additional IDs in this box once in the tool. Clicking (1) Search for related terms launches the search. All terms with SSSOM mappings to any of the search terms are shown in the results. The search term, search term definition, match method, match type, matched term, match term synonym(s) and match term definition are shown in the results to help the user select the terms to add to the search. Clicking (2) the box in the Add to Search selects terms to be added to the HMDC search. Users may add multiple terms. Clicking (3) Add IDs to HMDC Search adds all selected term IDs and any additional IDs entered in the search box to the HMDC search.



HMDC results: (1) Search using the MP term ID only returns only mouse results. (2) Search using MP and HPO term IDs returns both mouse and human results.



Searching by IDs gives a cleaner, more precise set of search results compared to text search which includes imprecise matches based on matches to labels and synonyms in the MP, HPO and DO.

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